

EECS 841: Computer Vision

Lecture 2

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Outline

- Recap about course
- Introduction to IP/CV
- Applications
- The problems “formally” solved in IP/CV
- Basics:
 - image definition
 - accessing image data
 - storing
 - types of images

Introduction

Course is about IP and CV

**What is your idea about image processing and computer vision?
Are they same or different concepts?**

Image Processing:

The processing of images using mathematical operations by using any form of signal processing for which the input is an image, a series of images or a video, such as a photograph or video frame; the output of image processing may be either an image or a set of characteristics or parameters related to the image. (Gonzales & Woods)

Computer Vision:

is an interdisciplinary field that deals with how computers can be made for gaining high-level understanding from digital images or videos. From the perspective of engineering, it seeks to automate tasks that the human visual system can do.

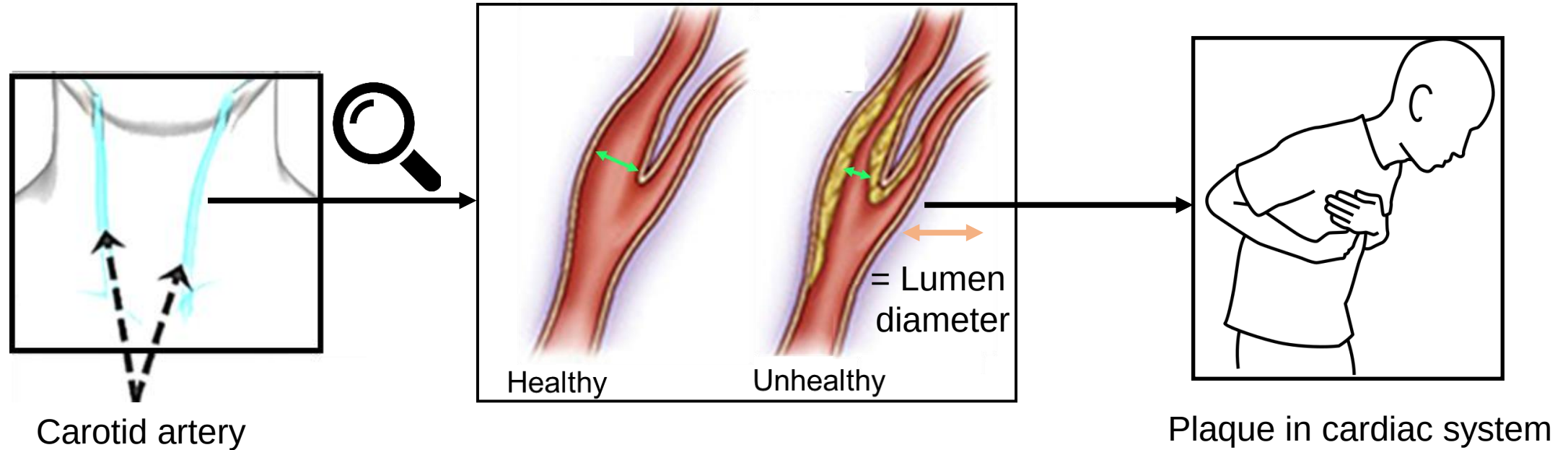
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Applications which use Image Processing / Computer Vision

What are the domains you think IP/CV is useful?
Name some applications?

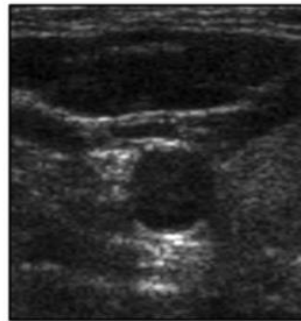
Healthcare: diagnosis



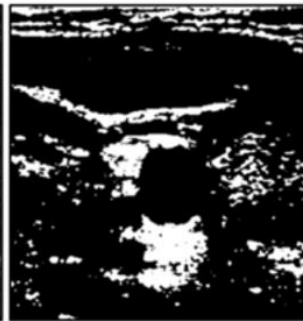
Main idea: *Measure carotid's lumen diameter to early detect cardiovascular plaque*

Anomaly Detection

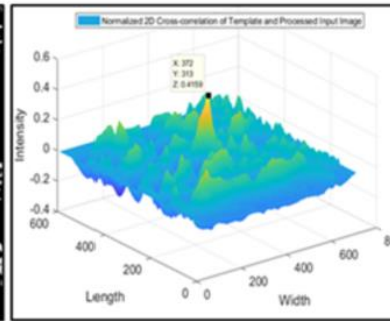
No plaque:



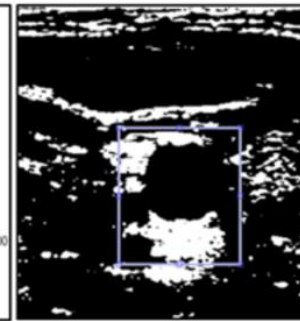
Raw image



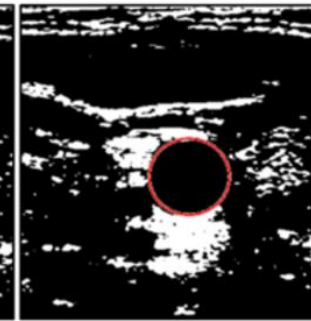
Processed image



Surface plot for detecting location of carotid



Detected location



Lumen diameter measurement

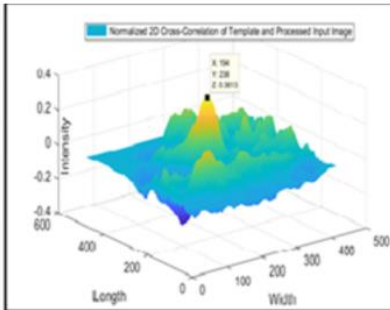
With plaque:



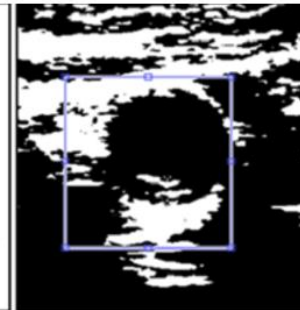
Raw image



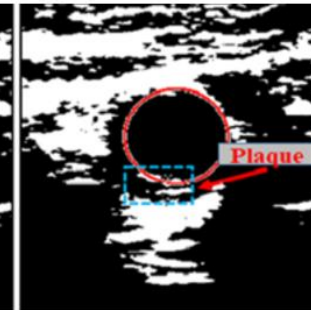
Processed image



Surface plot for detecting location of carotid



Detected location



Lumen diameter measurement

Sports: decision making

Video-based Tracking of a Baseball Pitch



from http://www-static.cc.gatech.edu/classes/AY2007/cs4495_fall/html/materials.html

- This is a snippet from a video feed
- A player is pitching
- The yellow box tells you where the ball should go in order to be considered as a strike

Result of baseball extraction



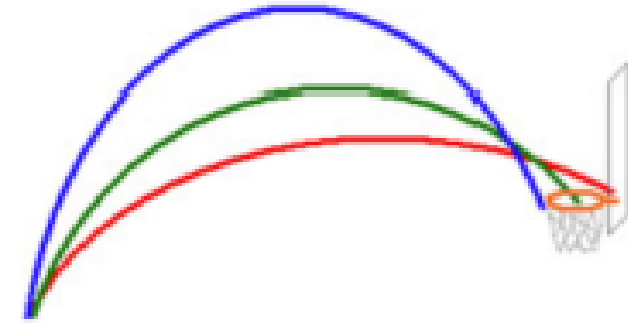
from http://www-static.cc.gatech.edu/classes/AY2007/cs4495_fall/html/materials.html

- Red lines show the actual pitch taken by the player
- IP/CV can precisely detect whether the ball after the pitch has been in the desired box to decide whether it is a strike or not

Sports: monitoring and improving performance



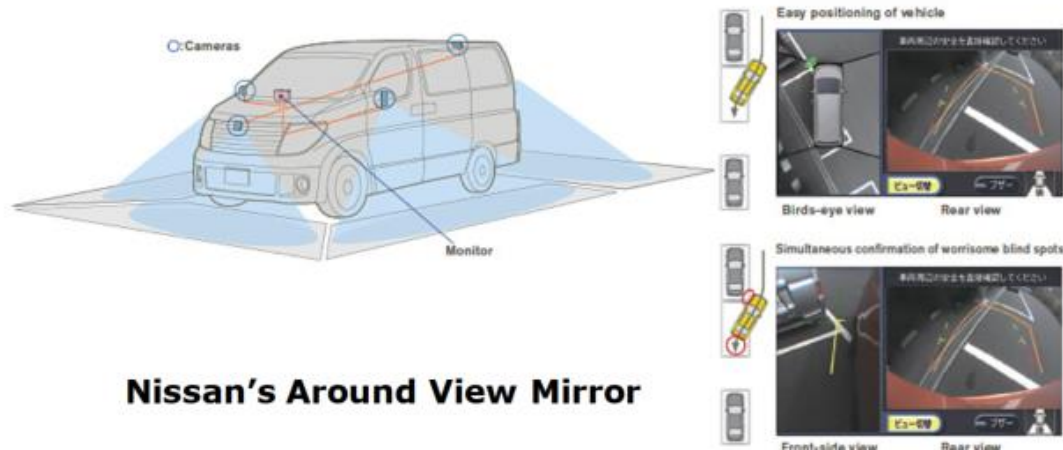
- CV/IP is useful for monitoring or observing one's performance
- Here is a snippet from a video frame of how a person is shooting during free throw in basketball



- The colorful lines show the actual pitch taken by the player
- IP/CV helps to draw the arc of the ball
- Arc is utilized for improving the performance

Automotive application

Structured Road Following in Autonomous vehicle



Nissan's Around View Mirror

<http://www.nissan-global.com/EN/TECHNOLOGY/INTRODUCTION/DETAILS/AVM/>

- Cameras all around the car
- See everything around the car
- IP/CV can help in making wise decision by analyzing the images
 - identify stop sign
 - estimate optimum parking position
 - speed control

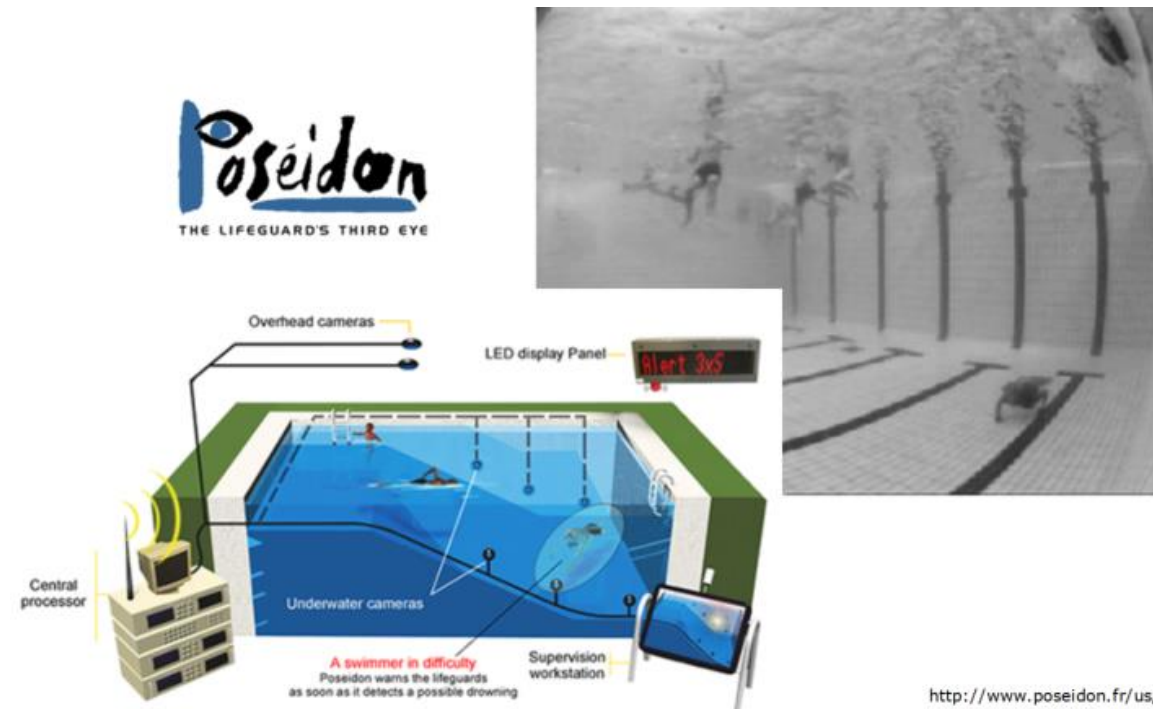
What if the road is unstructured?



from http://www-static.cc.gatech.edu/classes/AY2007/cs4495_fall/html/materials.html

- CV/IP can still help by object detection

Rescuing Drowning Victims



- Cameras installed under water
- They monitor whether a person has reached to certain depth or not
- If yes, then the behavior of the person is tracked: hand flapping/ facial expression unusual?
- If yes, send an alarm to guard

Special Effects in Movies

- *Bullet Time* special-effect in The Matrix is an example of the application of SFM ideas.
- Linear array of cameras replaces moving camera.



from http://www-static.cc.gatech.edu/classes/AY2007/cs4495_fall/html/materials.html

Surveillance

Tracking Multiple Humans



from http://www-static.cc.gatech.edu/classes/AY2007/cs4495_fall/html/materials.html

Behavior Analysis

Tracking Ants



from http://www-static.cc.gatech.edu/classes/AY2007/cs4495_fall/html/materials.html

Face Detection

Face Detection



Structure Creation from Motion

Structure from motion results

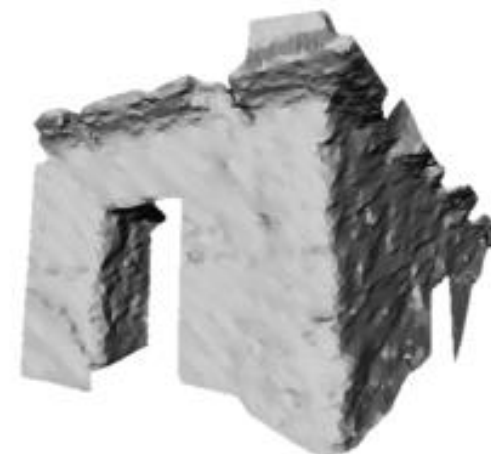
Input
frames
(3 of 5)



Texture-mapped model

Shaded model

Output
3-D model



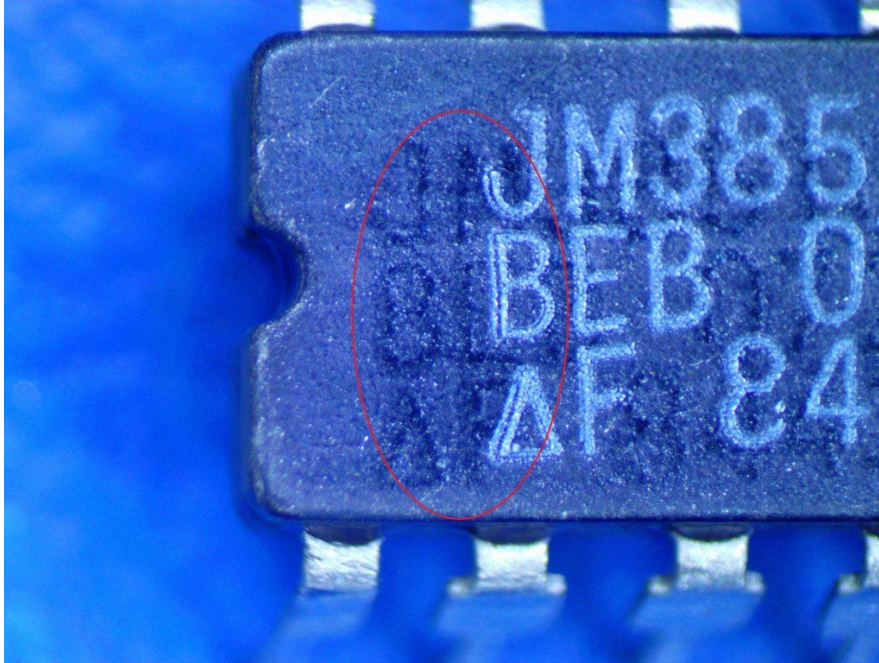
(Images courtesy of Luc Van Gool)

Forensics

- Crime scene recording and analysis



Hardware Security



- IP/CV useful for verify specifications
- Ghost marking are detected

- **IP/CV has been very useful in different domains for multi purpose activities!**
- **This course will teach you some of the basic ideas and a few of the latest concepts!**

Application Summary

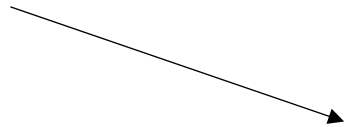
- Industrial inspection
- Document image analysis
- Transportation
- Security and surveillance
- Remote sensing
- Scientific imaging
- Medical imaging
- Robotics

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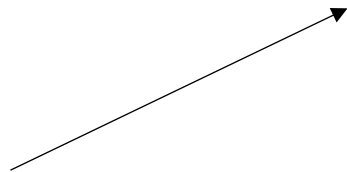
Definition of “Problems Solved” in the IP/CV

- **Image processing:** the field of study in which algorithms operate on input images to produce output images.



- **Image analysis:** the field of study in which algorithms operate on images to extract higher-level information.

(doing image processing = IP → IA → collect high level info = CV)



- **Computer vision:** refers to systems operating on images taken in unstructured settings, such as those taken by ordinary people in everyday life using their personal digital cameras.

Definition of “Problems Solved” in the IP/CV

The problems/tasks:

1. Enhancement
 - **Enhancement:** an image processing problem that involves transforming an input image into another image so as to improve its visual appearance.
2. Restoration
 - **Restoration:** an image processing problem that has as its purpose to restore an image that has been corrupted by some type of noise.
3. Compression
 - **Compression:** an image processing problem that involves storing an image with fewer bits than are required by the original signal.

Definition of “Problems Solved” in the IP/CV

The problems/tasks:

- 4. Segmentation
 - **Segmentation:** an image analysis problem that involves the process of determining which pixels in an image belong together, that is, which pixels are projections of the same object in the scene.
- 5. Classification
 - **Classification:** an image analysis problem that involves determining which pixels in an image belong to a model that has been created beforehand.
- 6. Shape from X
 - **Shape from X:** an image analysis problem that aims to recover the three-dimensional (3D) structure of the scene using any of a variety of techniques.

Quick Question

- **Machine vision:** refers to systems in an industrial setting in which the placement of the camera and lighting conditions can be controlled.
- **Computer vision:** refers to systems operating on images taken in unstructured settings, such as those taken by ordinary people in everyday life using their personal digital cameras.

For which scenario, the “Image Analysis” seems to be more challenging?

- A. Machine Vision
- B. Computer Vision

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Image Basics

- The word *image* comes from the Latin word (*imago*), meaning “likeness.”
- **Digital image:** the image captured by a camera is usually a digitized version of some two-dimensional sensory input.

Image Data

- **Digital image:** a discrete two-dimensional array of values, like a matrix.
 - *Width:* refers to the number of columns in the image.
 - *Height:* refers to the number of rows in the image.
- **Aspect ratio:** width divided by height.
- **Pixel:** Each element of the array.
- An image is stored in memory as a 1D array



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Image Data Accessing

If " I " is an image with h = height and w = width:
then a pixel from I can be accessed with $I(j,i)$ where j denotes a particular column and i denotes a particular row

Width = no. of columns = 5
($j = 0 : w-1$)

height = no. of rows = 3
($i = 0 : h-1$)

	j=0	1	2	3	4
i=0	1	2	3	4	5
1	6	7	8	9	10
2	11	12	123	14	15

Image Data Storing

- An image is stored in memory as a 1D array in two ways:
 - *Column major order*: the first column is stored, then the second column, and so on until the final column.
 - *Row major order*: refers to storing the first row, then the second row, and so on until the final row.

Width = no. of columns = 5

height = no. of rows = 3

1	2	3	4	5
6	7	8	9	10
11	12	123	14	15

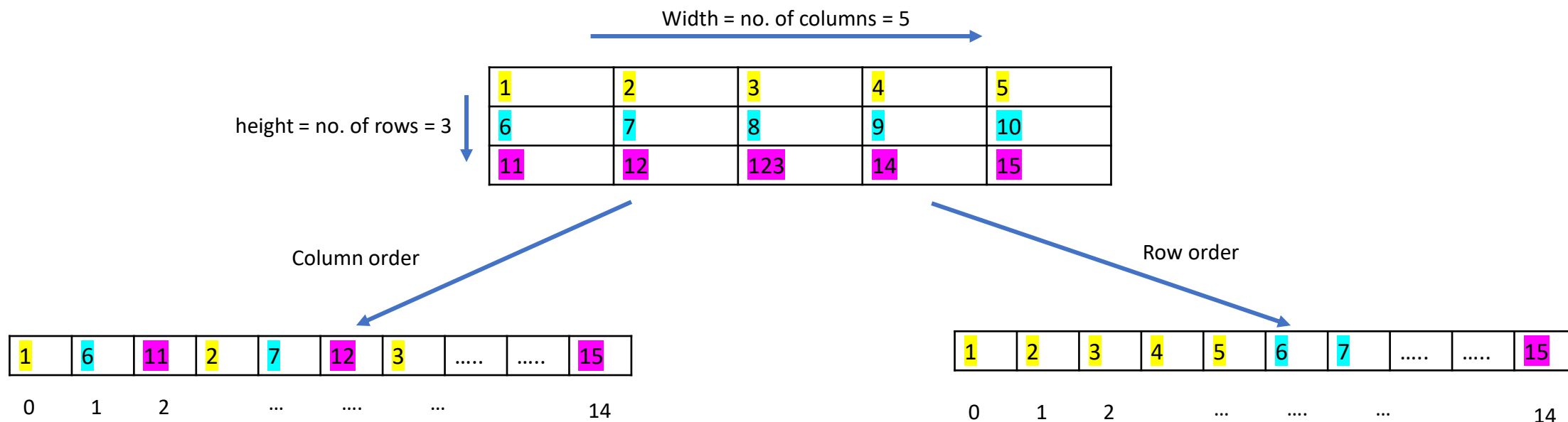
Q1: how many indices do we need in 1D array to store in memory?

Q2: what is the value of 5th index in that 1D array if you follow row order? (Consider the index begins from 0)

Q3: what is the value of 5th index in that 1D array if you follow column order? (Consider the index begins from 0)

Image Data

- An image is stored in memory as a 1D array in two ways:
 - *Column major order*: the first column is stored, then the second column, and so on until the final column.
 - *Row major order*: refers to storing the first row, then the second row, and so on until the final row.



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Types of Image

- **Grayscale image:** the value of each pixel is a scalar indicating the amount of light captured.
 - These values are quantized into a finite number of discrete levels called **gray levels**.
- In an 8-bit grayscale image, a pixel whose value is 0 represents black, whereas a pixel whose value is 255 represents white.

3	5	6
3	2	3
1	9	1

Is it a
dark/bright
image?

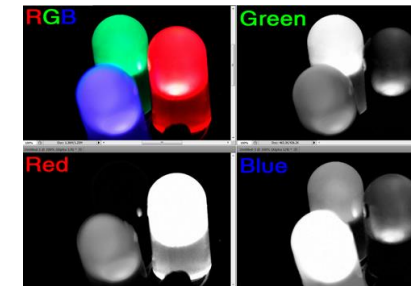
“ $I(x,y)$ returns one value in
grayscale image
whereas 3 values are returned
in RGB images”

- **RGB color image:** the pixel values are triples containing the amount of light captured in the three color channels: red, green, and blue.

Types of Image: A bit more clarification...

“ $I(x,y)$ returns one value in grayscale image whereas 3 values are returned in RGB images”

- $I(x,y) = (7,8,112) \rightarrow$ means the amount of intensity in the red, green, and blue channels is 7,8, and 112 respectively at the image's x,y location
- The range of intensity in each channel can be: $[0: 255]$
- If you check only one channel individually, it will look like a gray image but depending on whether you were watching it on R or G or B channel, the grayness will be different $\rightarrow$ looking at them all together gives you the color image



- $I(x,y) = 15 \rightarrow$ means it is the amount of intensity in one channel $\rightarrow$ but which one? R/G/B? It can be coming from any of the three or the average of the three or any other value derived from any composition technique from the three channels.

Storage of RGB Image

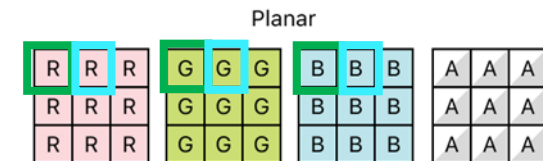
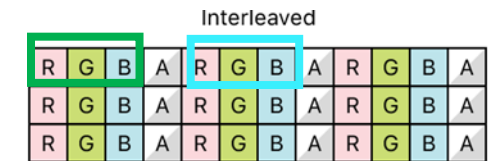
- **RGB color image:** the pixel values are triples containing the amount of light captured in the three color channels: red, green, and blue.
- **Interleaved:** all three values for one pixel are stored before the three values of the next pixel
- **Planar:** the red, green, and blue channels are stored as separate one-byte-per-pixel images
- **Alpha value or opacity:** used for blending multiple images



	j=0	1	2
i=0	(0,0)	(1,0)	(2,0)
1	(0,1)	(1,1)	(2,1)
2	(0,2)	(1,2)	(2,2)

https://developer.apple.com/documentation/accelerate/optimizing_image-processing_per

A = alpha = amount of transparency you want to give to the color
It ranges from 0 to 1 where 0 being fully transparent and 1 being not transparent (=opaque)



Note: according to the book, when expressing the pixel location of an Image as $I(x,y)$, we put the column first (in x position) and the row in the second place (in y position)

Types of Images: More types...

- **Binary image:** The logical values can be stored using one bit per pixel, (0 for off or 1 for on)
- **Real-valued image, or floating-point image:** each pixel contains a real number.
 - The number is stored in the computer as an IEEE single- or double-precision floating point number
- **Integer-valued image:** the value of each pixel is an integer.
- **Complex-valued image:** arises from computing the Fourier transform of an image.
 - It contains two floating-point values for each pixel, one for the real component and one for the imaginary component.

Summary of Different Image types

	grayscale	RGB color	binary	integer-valued	real-valued	complex-valued
channels	1	3	1	1	1	2
bit depth	8	24	1	32/64	32/64	64/128
value range	$\{0, \dots, 255\}$	$\{0, \dots, 255\}^3$	$\{0, 1\}$	$\mathbb{Z}$	$\mathbb{R}$	$\mathbb{R}^2$

TABLE 1.3: Common image types, shown with the number of channels, the most commonly encountered bit depth (number of bits per pixel), and the set of possible values. In the final three columns this set is conceptual only, since the integers $\mathbb{Z}$ and real numbers $\mathbb{R}$ are infinite sets.